

AI Applications Lecture 5

Pipelines and Tokenization

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^aI thank my collaborator, Jing WANG, for cooperating in creating the teaching materials.

Outline

Introduction

How does a NN work in generative AI? - Pipeline

What do we expect on a natural language generation pipeline? - Interface

Tokenization

Subword Tokenization: Byte Pair Encoding (BPE)

Special Tokens and Chat AI

Conclusion and Future Outlook

Introduction

Review of the previous lecture

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- We confirmed the non-trivial fact that the separation of **architecture** and **checkpoints** is crucial for understanding the scope of license application.
- We also saw that even models touted as "open" are not necessarily permitted for free use.
- In the exercise, we experienced the process of downloading a model from an online repository like the Hugging Face Hub, checking its license, and then using it.

Learning outcomes of this lecture

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- Explain the significance and benefits of using a **pipeline**, which combines a neural network with other functions, in the practical application of generative AI.
- Given a pipeline provided by Hugging Face, roughly understand its structure as a function (algorithm) from its source code.
- Convert natural language strings into sequences of **tokens** for input into a neural network, and conversely, convert token sequences generated by the neural network back into natural language strings.

How does a NN work in generative AI? - Pipeline

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Question: A neural network is just a continuous function. How does it work as a component in a real application?

Neural networks and pipelines

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In most cases, **one or more neural networks** are combined with **other non-neural network algorithms (functions)** to build a larger single function (system).

In this lecture, we will refer to this entire composite function as a **pipeline**.

Neural networks and pipelines

Remark

The term "pipeline" is used as a central concept, particularly in libraries developed by Hugging Face such as `transformers`¹ and `diffusers`². While this term is less frequently used with the same meaning in theoretical academic literature, it is a very useful concept for understanding practical AI systems.

¹https://huggingface.co/docs/transformers/en/pipeline_tutorial

²https://huggingface.co/docs/diffusers/using-diffusers/write_own_pipeline

Neural networks and pipelines

The purposes of constructing a pipeline are diverse, but the main motivations are as follows:

1. To reduce the amount of data handled by the core neural network.

- **Example 1: Tokenizer in Chat AI.** It converts long sentences into shorter sequences of tokens, reducing the sequence length that the neural network has to process. (The main topic of this lecture)
- **Example 2: VAE in Image Generation.** It compresses high-resolution images into low-dimensional latent vectors, so expensive models only operate in this small space.

Neural networks and pipelines

The second reason is:

To interconvert between discrete objects (e.g., strings) and continuous objects (real-valued vectors) that neural networks excel at.

- **Example 1 (overlapping with above): Tokenizer in Chat AI.** It converts strings into sequences of numbers (token IDs).
- **Example 2: Sampling in Chat AI.** It selects a specific, discrete token from a continuous probability distribution output by the network.

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- **Training:** Training data and one or more neural network architectures are given to a training algorithm to obtain **checkpoints** corresponding to each architecture.
- **Inference:** A **pipeline** ϕ , constructed by combining one or more trained models and non-neural network functions, is given a new input to obtain the final output.

$$\phi_{f_{\theta^{(1)*}, \dots, g^{(1)}, \dots}}^{(1)} : \mathcal{X} \rightarrow \mathcal{Y}$$

Updating the framework of training and inference

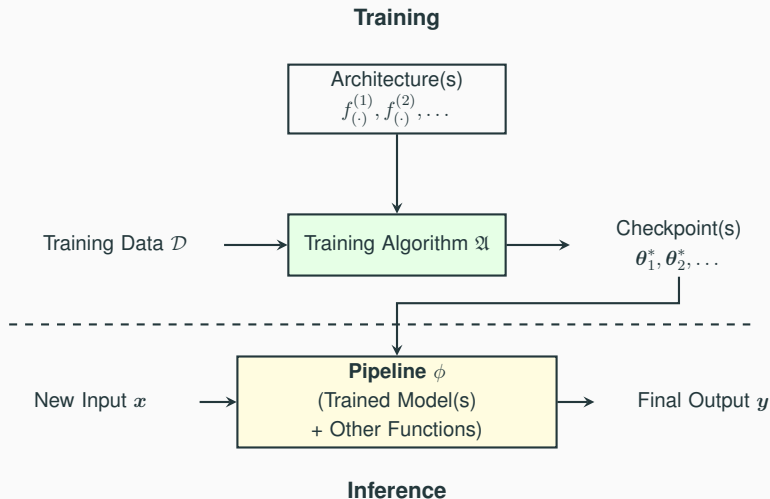


Figure 1: Scheme of Training and Inference Including a Pipeline

How to understand the structure of a pipeline

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Therefore, reading the implementation code of libraries like Hugging Face is an effective way to grasp the overall picture. However, these libraries are huge, so where do you start?

How to understand the structure of a pipeline

Here is a recommended procedure for grasping the overview of a pipeline:

1. **Identify the pipeline's class:** First, check the class name of the instance loaded with a method like `from_pretrained` (e.g., `print(type(pipeline_object))`).

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3. **Check the execution method** (`__call__`): Trace the processing of the class's `__call__` method. This will show you how the components are combined and how the series of processes from input to output is executed.

Question: How

Question: How does a NN work as a component in a real application?

Answer: Neural networks are composed with other functions and work as a parametric function. The composite function is called a **pipeline**.

**What do we expect on a natural
language generation pipeline? -
Interface**

Question: What structure does the pipeline take in natural language generation AI?

Structure of a natural language generation pipeline

Let's take the natural language generation task as a specific example. A typical pipeline used in auto-regressive language models can be broadly seen as a composite function of the following components.

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Structure of a natural language generation pipeline

Let's take the natural language generation task as a specific example. A typical pipeline used in auto-regressive language models can be broadly seen as a composite function of the following components.

1. **Tokenizer / Encoder:** Converts a natural language string into a sequence of integer IDs (tokens).
2. **Neural Network and Sampler:** Takes a token sequence and returns another token sequence.
3. **Detokenizer / Decoder:** Converts the generated token sequence back into a natural language string.

The overall picture of the pipeline

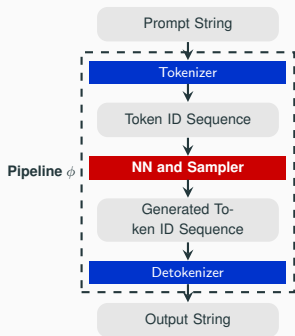


Figure 2: Detailed Inference pipeline.

Mathematically, the entire pipeline ϕ is a composition of these functions:

$$\phi = \text{Detokenizer} \circ \text{Sampler}_{f_{\theta^*}} \circ \text{Tokenizer}$$

Since it contains the parametric function f_{θ^*} , the entire pipeline is also a parametric function dependent on the checkpoint θ^* .

Structure of a natural language generation pipeline

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This is because it is difficult to correctly understand the behavior of the neural network and sampler inside without understanding tokenization.

Natural language generative AI pipeline - answer

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Question: What structure does the pipeline take in natural language generation AI?

Answer: The composition of the following:

- Tokenizer (encoder),
- Neural network,
- Sampler,
- Detokenizer (decoder)

Tokenization

Question: How does the natural language generation pipeline convert raw characters to an input number sequence?

The necessity of tokenization

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The larger a neural network's architecture, the more complex functions it can represent, and performance improvement can be expected.

Recent high-performance models have a vast number of parameters (e.g., Llama 3 70B, Qwen2 72B).

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However, the giantization of architecture comes with serious disadvantages.

- **Merit:** If the checkpoint is chosen appropriately, a complex function can be obtained, and high performance can be expected.
- **Demerit:** The computational resources (memory, computing power) and time required for training and inference become enormous. This limits where they can be used and increases costs.

The necessity of tokenization

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Pipelining is an important strategy to solve this problem.

By sandwiching relatively lightweight processes (like tokenization) before and after the neural network, the burden on the network itself is reduced, achieving a balance between efficiency and performance.

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The reason is that **the proximity of byte values is completely unrelated to the proximity of the meanings they represent.**

For example, in UTF-8, 'A' ('0x41') and 'B' ('0x42') are numerically close. But 'a' ('0x61') is numerically distant from 'A'. Neural networks, being continuous functions, would incorrectly infer that 'A' and 'B' are similar, while 'A' and 'a' are very different.

Why not input byte streams directly?

Remark

This situation is in contrast to image data. An image is represented by a tensor of pixel color information. The fact that pixel values are close means that the colors are physically close, which often leads to semantic closeness (such as the surface of the same object). Therefore, it is common for the pixel data of an image to be normalized and then input directly into a neural network.

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It is difficult to decide in advance "which characters are semantically close," and it could introduce human bias.

Therefore, the simplest solution is to **assume that all characters are equidistant**. The idea that embodies this is **One-Hot Encoding**.

One-hot encoding

Definition (One-Hot Encoding)

Let a finite vocabulary set $\mathcal{V} = \{v_1, v_2, \dots, v_D\}$ be given, where $D = |\mathcal{V}|$ is the vocabulary size. The One-Hot encoding of an element $v_i \in \mathcal{V}$ is a D -dimensional vector $e_i \in \{0, 1\}^D$ in which only the i -th component is 1 and all other components are 0.

$$(e_i)_j = \begin{cases} 1 & \text{if } j = i \\ 0 & \text{if } j \neq i \end{cases} \quad (1)$$

One-hot encoding

Example (One-Hot Encoding of Characters)

Let the vocabulary be $\mathcal{V} = \{a, b, c\}$.

- 'a' $\rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
- 'b' $\rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$
- 'c' $\rightarrow \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

In this representation, the Euclidean distance between any two vectors is $\sqrt{2}$, and equidistance is maintained.

Motivation for shortening sequence length

When each character is converted into a One-Hot vector, a sentence becomes a sequence of vectors. For example, "abac" becomes a sequence of four 3D vectors.

$$\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right)$$

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Now, consider making the unit of conversion larger than a character, for example, a word. If we choose words as our vocabulary, the sequence length becomes dramatically shorter.

Motivation for shortening sequence length

Example (Character-level vs. Word-level)

Sentence: "Language models are powerful."

- **Character-level (28 characters):** 'L', 'a', 'n', 'g', 'u', 'a', 'g', 'e', ' ', ' ', 'm', ...
- **Word-level (4 words):** "Language", "models", "are", "powerful"

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For simplicity, converting each unit to an integer ID would look like this:

- **Character-level:** (44, 65, 78, ..., 76) (length 28)
- **Word-level:** (1345, 11646, 322, 2) (length 4)

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Using words as the basic unit creates new problems.

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- **Out-of-Vocabulary (OOV) Problem:** Cannot handle words not in the vocabulary (new words, proper nouns, typos).
- **Vocabulary Explosion:** Trying to cover many words results in a huge vocabulary size, increasing the dimensionality of the One-Hot vectors.
- **Difficulty in Multilingual Support:** It is necessary to have a huge vocabulary for each language, which is inefficient.

Tokenization - answer

Question: How does the natural language generation pipeline convert raw characters to an input number sequence?

Answer: Each discrete entity in the input sequence is converted to a one-hot vector.

Subword Tokenization: Byte Pair Encoding (BPE)

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Question: In English, a letter does not have a semantic meaning. On the other hand, in Chinese, a character has its own semantic meaning. It is not trivial what should be the minimum unit in natural language sentences.

What should be the “unit” of tokenization? How can we automatically determine the unit?

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The basic idea of BPE is to build a data-adapted vocabulary by **merging frequently occurring pairs of characters into a new single unit (token)**.

The BPE algorithm

BPE is divided into two phases:

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1. A **dictionary creation phase** that builds the vocabulary and merge rules.
2. An **encoding/decoding phase** that uses them to process text.

The BPE algorithm

Definition (BPE Dictionary Creation Algorithm)

The inputs are a training corpus $\mathcal{C}$ and the desired number of merges N .

1. Initialization:

- Initialize the vocabulary $\mathcal{V}$ with all unique characters in $\mathcal{C}$.
- Split all text in the corpus into sequences of these characters.
- Initialize an empty, ordered list of merge rules m .

The BPE algorithm

Definition (BPE Dictionary Creation Algorithm (cont.))

2. **Iteration:** For $k = 1$ to N , repeat the following:
 - 2.1 Count the frequency of all adjacent token pairs in the corpus.
 - 2.2 Find the most frequent pair (t_a, t_b) .
 - 2.3 Generate a new token t_{new} by concatenating t_a and t_b .
 - 2.4 Add the new token t_{new} to the vocabulary $\mathcal{V}$.
 - 2.5 Add the rule $(t_a, t_b) \rightarrow t_{\text{new}}$ to our list of merge rules m .
 - 2.6 Replace all occurrences of the pair (t_a, t_b) in the corpus with t_{new} .

The outputs are the final vocabulary $\mathcal{V}$ and the ordered merge rules m .

The BPE algorithm

Definition (BPE Encode/Decode Algorithms)

- **Encoding:** Given a string and the ordered merge rules from dictionary creation.
 1. Split the string into a sequence of characters.
 2. For each merge rule, in the order they were created, replace all corresponding adjacent pairs in the sequence with the new token.
 3. Output the final token sequence.

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 3. Output the final token sequence.
- **Decoding:** Given a token sequence.
 1. Concatenate all tokens in order to form a single string.
 2. Output the string.

The BPE algorithm

Example (Detailed Example of BPE Operation)

Suppose the training corpus is just: "fast faster fastest slow slower slowest". We'll treat words as being separated.

1. Initial State:

- Vocabulary $\mathcal{V}$: 'f', 'a', 's', 't', 'e', 'r', 'l', 'o', 'w'
- Data $\mathcal{D}$: 'f a s t', 'f a s t e r', 'f a s t e s t', 's l o w', 's l o w e r', 's l o w e s t'

The BPE algorithm

Example (Detailed Example of BPE Operation (cont.))

2. 1st Merge:

- We count all adjacent pairs. The pair '(s, t)' appears 6 times, which is the most frequent.
- We create a new token 'st' and add it to the vocabulary.
- Our data becomes: 'f a st', 'f a st e r', 'f a st e st', 's l o w', 's l o w e r', 's l o w e st'

The BPE algorithm

Example (Detailed Example of BPE Operation (cont.))

3. 2nd Merge:

- The pairs '(f, a)' and '(s, l)' are now the most frequent (3 times each). We'll pick '(f, a)' based on alphabetical order.
- New token: 'fa'. Data: 'fa st', 'fa st e r', 'fa st e st', ...

The BPE algorithm

Example (Detailed Example of BPE Operation (cont.))

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- New token: 'fa'. Data: 'fa st', 'fa st e r', 'fa st e st', ...

4. 3rd Merge:

- The most frequent pair is now '(fa, st)' (3 times).
- New token: 'fast'. Data: 'fast', 'fast e r', 'fast e st', ...

The BPE algorithm

Example (Detailed Example of BPE Operation (cont.))

By repeating this process, frequent words like 'slow' and frequent suffixes like 'er' and 'est' are learned as single tokens.

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This allows an unknown word like 'slowing' to be handled by splitting it into the learned 'slow' and the unknown 'ing', partially preserving information.

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Example (Detailed Example of BPE Operation (cont.))

By repeating this process, frequent words like 'slow' and frequent suffixes like 'er' and 'est' are learned as single tokens.

This allows an unknown word like 'slowing' to be handled by splitting it into the learned 'slow' and the unknown 'ing', partially preserving information.

Also, by starting BPE from **bytes** instead of characters, the complexity of Unicode can be avoided, and all languages and emojis can be handled uniformly.

The BPE algorithm

Exercise

Suppose the corpus is “unzip”, “unzipped”, “zip” and the basic units are “u”, “n”, “z”, “i”, “p”, “e”, “d”.

If you perform 5 merges, what will the final merge rules and vocabulary be?

The BPE algorithm

Solution:

1. **Initial State:** Data is 'u n z i p', 'u n z i p p e d', 'z i p'.

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2. **1st Merge:**
 - Most frequent pairs: '(z, i)' and '(i, p)' (3 times each). Pick '(i, p)' alphabetically.
 - Rule 1: 'i p' $\rightarrow$ 'ip'.
 - Data: 'u n z ip', 'u n z ip p e d', 'z ip'.

The BPE algorithm

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 - Rule 1: 'i p' $\rightarrow$ 'ip'.
 - Data: 'u n z ip', 'u n z ip p e d', 'z ip'.
3. **2nd Merge:**
 - Most frequent pair: '(z, ip)' (3 times).
 - Rule 2: 'z ip' $\rightarrow$ 'zip'.
 - Data: 'u n zip', 'u n zip p e d', 'zip'.

The BPE algorithm

Solution (cont.):

3. 3rd Merge:

- Most frequent pairs: '(u, n)' and '(n, zip)' (2 times each). Pick '(n, zip)'.
- Rule 3: 'n zip' $\rightarrow$ 'nzip'. Data: 'u nzip', 'u nzip p e d', 'zip'.

The BPE algorithm

Solution (cont.):

3. 3rd Merge:

- Most frequent pairs: '(u, n)' and '(n, zip)' (2 times each). Pick '(n, zip)'.
- Rule 3: 'n zip' $\rightarrow$ 'nzip'. Data: 'u nzip', 'u nzip p e d', 'zip'.

4. 4th Merge:

- Most frequent pair: '(u, nzip)' (2 times).
- Rule 4: 'u nzip' $\rightarrow$ 'unzip'. Data: 'unzip', 'unzip p e d', 'zip'.

The BPE algorithm

Solution (cont.):

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- Most frequent pairs: '(u, n)' and '(n, zip)' (2 times each). Pick '(n, zip)'.
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- Rule 4: 'u nzip' $\rightarrow$ 'unzip'. Data: 'unzip', 'unzip p e d', 'zip'.

5. 5th Merge:

- Many pairs occur once. Pick '(e, d)' alphabetically.
- Rule 5: 'e d' $\rightarrow$ 'ed'. Data: 'unzip', 'unzip p p ed', 'zip'.

The BPE algorithm

Exercise

Use the rules you just created to encode the string "unzipping".

Final Result and Encoding:

- **Merge rules (ordered):** '(i, p)', '(z, ip)', '(n, zip)', '(u, nzip)', '(e, d)'
- **Encoding "unzipping":**
 1. Initial: ['u', 'n', 'z', 'i', 'p', 'p', 'i', 'n', 'g']
 2. After rule 1 ('ip'): ['u', 'n', 'z', 'ip', 'p', 'i', 'n', 'g']
 3. After rule 2 ('zip'): ['u', 'n', 'zip', 'p', 'i', 'n', 'g']
 4. After rule 3 ('nzip'): ['u', 'nzip', 'p', 'i', 'n', 'g']
 5. After rule 4 ('unzip'): ['unzip', 'p', 'i', 'n', 'g']
 6. Rule 5 is not applicable.
- **Final token sequence:** ['unzip', 'p', 'i', 'n', 'g']

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- The neural network receives this token ID sequence and outputs a sequence of probability distributions for the next token.
- The sampler selects the next token ID from the probability distribution.
- BPE **decoding** converts the generated token ID sequence into a byte sequence corresponding to natural language.

The role of BPE in the pipeline

Remark

In the case of byte-level BPE, there is no guarantee that the decoded byte sequence will constitute a valid UTF-8 string, etc. It is possible for invalid byte sequences to be generated.

Subword-tokenization

Question: What should be the “unit” of tokenization? How can we automatically determine the unit? **Answer:** We regard a frequently appearing subsequence of letters as “token” and convert it to a one-hot vector. We automatically determine them using an information-theoretic algorithm.

Special Tokens and Chat AI

Question: In Chat AI, text is not everything. We also implicitly input past conversations or system prompts.

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How do we let AI input/output such complex data?

Special tokens and Chat AI

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However, the implementation of widely used Chat AIs today is insufficient with just this. This is because the input to a Chat AI is not a single string, but includes structural information.

Special tokens and Chat AI

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- **Special Tokens** are tokens used to represent the structure of the conversation (e.g., who is speaking, start/end of a message).
- A **Chat Template** is a set of rules for converting a structured conversation history into a single string containing these special tokens, which can then be fed to the tokenizer.

Chat templates and special tokens

The input to a Chat AI is often given as a list of dictionaries recording "who said what."

```
1 messages = [  
2     {"role": "system", "content": "This is a test. Think as short as possible."  
3     },  
4     {"role": "user", "content": "Find 7+8."},  
5 ]
```

Chat templates and special tokens

This structured data needs to be converted into a single string. The Hugging Face tokenizer's `apply_chat_template` method does this. For the Qwen model, the previous 'messages' are converted into a string like this:

```
<|im_start|>system
```

```
This is a test. Think as short as possible.<|im_end|>
```

```
<|im_start|>user
```

```
Find 7+8.<|im_end|>
```

```
<|im_start|>assistant
```

```
<think>
```

Chat templates and special tokens

This string contains special tokens in addition to the normal BPE vocabulary:

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Chat templates and special tokens

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- `<|im_start|>` and `<|im_end|>`: Mark the start and end of an utterance.
- `system`, `user`, `assistant`: Explicitly state the role of the utterance.
- `<think>` and `</think>`: Mark the start and end of the model's internal "thinking" process.

Chat templates and special tokens

Particularly important is the part at the end: `<|im_start|>assistant<think>`.

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The language model's task is to complete a given token sequence. Models that use "Thinking" are trained on complete sequences of the form:

"Question <think> Reasoning </think> Answer"

Chat templates and special tokens

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The language model's task is to complete a given token sequence. Models that use "Thinking" are trained on complete sequences of the form:

"Question <think> Reasoning </think> Answer"

Therefore, by giving *"Question <think>"* as input, the model is prompted to generate the remaining part: *"Reasoning </think> Answer"*.

Chat templates and special tokens

Remark

We have stated that the goal of AI is to acquire a function that represents an appropriate input-output relationship. For a Chat AI, it seems sufficient to take a question as input and output a response.

Chat templates and special tokens

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We have stated that the goal of AI is to acquire a function that represents an appropriate input-output relationship. For a Chat AI, it seems sufficient to take a question as input and output a response.

However, large language models (LLMs) used in Chat AI have ended up learning a **token sequence completion function**.

Chat templates and special tokens

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The difference between a "question-to-answer" function and a "token sequence completion" function lies in their domains. The completion function has an unnecessarily larger domain.

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Chat templates and special tokens

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The difference between a "question-to-answer" function and a "token sequence completion" function lies in their domains. The completion function has an unnecessarily larger domain.

It accepts not just a "*Question*", but also inputs like "*Question Beginning of the answer*". This allows one to control the direction of the response.

This property is both convenient and can be used for attacks, such as Prefix Injection or Prefilling Attacks [3, 4].

Question: How do we let AI input/output such complex data, including past conversations and system prompts?

Answer: We flatten those complex data into a token sequence using special tokens. AI handle those flattened sequences.

Conclusion and Future Outlook

The overall picture of the pipeline

Integrating the discussions so far, let's reconfirm the overall picture of the pipeline. It is a composition of a series of functions:

1. **Template Renderer** (Renderer): Converts structured 'messages' into a single prompt string.

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1. **Template Renderer (Renderer)**: Converts structured 'messages' into a single prompt string.
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3. **Neural Network and Sampler (Sampler _{f_{θ^*}})**: Takes a token ID sequence and returns another. This is the core parametric function.
4. **Detokenizer (Decoder) (Detokenizer)**: Converts the generated token IDs back into a human-readable string.

The overall picture of the pipeline

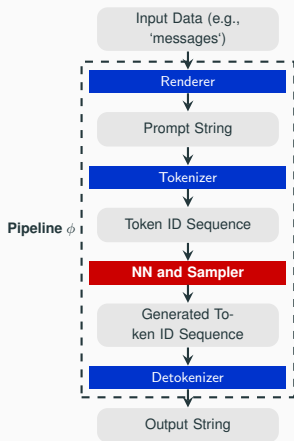


Figure 3: Detailed Inference pipeline.

Mathematically, the entire pipeline ϕ is a composition of these functions:

$$\phi = \text{Detokenizer} \circ \text{Sampler}_{f_{\theta^*}} \circ \text{Tokenizer} \circ \text{Renderer}$$

Since it contains the parametric function f_{θ^*} , the entire pipeline is also a parametric function dependent on the checkpoint θ^* .

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Summary of today's lecture

- In practical AI, a **pipeline** that combines multiple models and algorithms is used. This is a key strategy for balancing efficiency and performance.
- An important component in NLP pipelines is **tokenization**, which handles the conversion between human-readable strings and numerical data for the network.
- **Byte Pair Encoding (BPE)** is an effective subword tokenization algorithm that can handle unknown words while keeping sequence lengths manageable.

Summary of today's lecture

- In practical AI, a **pipeline** that combines multiple models and algorithms is used. This is a key strategy for balancing efficiency and performance.
- An important component in NLP pipelines is **tokenization**, which handles the conversion between human-readable strings and numerical data for the network.
- **Byte Pair Encoding (BPE)** is an effective subword tokenization algorithm that can handle unknown words while keeping sequence lengths manageable.
- In Chat AI, **special tokens** and **chat templates** are used to represent conversational structure and control the model's behavior.

Preview of the next lecture

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This time, we focused on tokenization, which handles the input to the neural network.

Next time, we will look in detail at the output side of the neural network, namely the technique of **sampling**, which is "how the next token is determined from a probability distribution."

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Safety alignment should be made more than just a few tokens deep.

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